

TOPIC 01: WHAT IS NODE.JS?

Definition:

Node.js is a **JavaScript Runtime Environment** that allows developers to execute JavaScript code outside the browser on the server-side.

CHROME V8 JS ENGINE

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C++ BINDINGS & LIBUV

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NODE.JS RUNTIME



Key Rule: Node.js is **NOT** a programming language, library, or framework! It is a runtime execution environment.

Node REPL: Typing node in the terminal launches the interactive Read-Evaluate-Print-Loop to test JS code directly.

TOPIC 02: PROCESS & PROCESS.ARGV

What is process?

process is a global object providing information about and control over the current running Node.js process.

process.argv (Command Line Arguments):

An array holding all arguments passed via the terminal when launching the Node.js application:

```
$ node app.js Shivam 22
```

Index	Value	Description
argv[0]	Path to Node.exe	Absolute filesystem path to Node executable
argv[1]	Path to app.js	Path to the script file being executed
argv[2]	"Shivam"	First custom user command-line argument
argv[3]	"22"	Second custom user argument

TOPIC 03: MODULE.EXPORTS & REQUIRE()

- `require()` → Built-in CommonJS function used to import external modules and files.
- `module.exports` → Special object used to expose functions, objects, or variables to other files.

MATH.JS: MODULE.EXPORTS = SUM; → `require('./math')` →

APP.JS: CONST SUM = REQUIRE(...)

Requiring a Directory (Folder):

When importing an entire directory: `const fruits = require('./fruits');` Node.js automatically searches for and executes the `index.js` file inside that folder!



Key Takeaway: Just like `main()` is the entry point in languages like Java, `index.js` serves as the default entry point for folder imports in Node.js!

TOPIC 04: NPM (NODE PACKAGE MANAGER)

NPM is the standard package manager for Node.js, fulfilling two core roles:

1. ONLINE REGISTRY (OPEN-SOURCE PACKAGE LIBRARY)

2. CLI TOOL (TERMINAL PACKAGE MANAGEMENT)

```
$ npm install <package_name> # e.g. npm i express
```

TOPIC 05: PACKAGE FILES & FOLDERS

File / Directory	Description & Purpose
node_modules/	Directory storing all downloaded packages and transitive sub-dependencies (added to .gitignore due to large size).
package.json	Holds project metadata (project name, version, start scripts, list of direct dependencies). Generated via <code>npm init -y</code> .
package-lock.json	Locks and records the exact version of every installed dependency and nested sub-dependency for reproducible builds.

TOPIC 06: LOCAL VS GLOBAL PACKAGES

- **Local Installation:** `npm i <package>` → Installed inside the local project's `node_modules`.
- **Global Installation:** `npm i -g <package>` → Available system-wide as CLI utilities (e.g. `nodemon`).
- **npm link:** Links a globally installed package to the local project without re-downloading.

TOPIC 07: REQUIRE VS IMPORT

Feature	<code>require()</code> [CommonJS]	<code>import</code> [ES6 Modules]
Syntax	<code>const mod = require('./mod')</code>	<code>import { mod } from './mod.js'</code>
Selective Loading	Loads entire file (No tree-shaking)	Selectively loads only required pieces (Saves Memory! ✅)
Loading Mode	Synchronous (blocking)	Asynchronous (non-blocking)
Configuration	Default in Node.js	Requires "type": "module" in package.json

